Anjum Ulazim

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Education

Indian Institute of Technology Guwahati, India Nov 2021 - July 2025
Bachelor of Technology

Experience

AI Engineer-I Sep 2025 – Present
SimplAI

- Integrated and extended a **Python FastAPI workflow orchestration** engine with step-level retries, fallback routing, and dynamic DAG execution, reliably supporting **50,000+ jobs/day** with workflow state persisted in **MongoDB**.
- Developed an **asynchronous OCR microservice** (FastAPI) integrating multiple OCR providers via a pluggable architecture, sustaining **180–200 pages/min throughput** while reducing document processing latency by **35%** overall.
- Engineered a production **AI guardrail framework with 12 validators** across **4 validation surfaces**, using **gRPC server/client streaming, request serialization, and validator/model reuse** for efficient validation of static and streaming LLM responses.
- Implemented an **event-driven Outlook email automation pipeline** using Microsoft Graph webhooks, managing subscription validation and renewals to process **1,000+ incoming email events/day** and trigger downstream AI workflows.
- Integrated a **voice automation module** into the platform, handling call initiation, lifecycle webhooks, transcripts, recordings, and pricing logic to support **100+ automated calls/day** with persistent call state tracking in MongoDB.

AI Engineering Intern March 2025 – Aug 2025
Sutra Sphere LLC

- Built backend APIs and frontend interfaces to fetch and display customer data during live calls, enabling real-time agent context retrieval across 500+ user records and reducing information lookup time by 30% for support agents.
- Implemented a **user invitation and admin approval system** with role-based access control, **validating 30+ conversational workflows** and identifying pipeline failures across STT, LLM, and TTS stages within the voice automation system.
- Contributed to development and testing of a **voice-based conversational AI agent** using Twilio with STT, LLM inference, and TTS pipelines, supporting **500+ automated calls/day** across multi-step conversation flows.

Projects

ClassyDiffusion Jan 2024 - Mar 2024
Personal Project

- Built a diffusion model (64M+ parameters, 246 MiB) using **DDPM with UNet + Attention Mechanism** and CIFAR-10 dataset to generate class-conditional images of 4 classes (airplane, automobile, bird, cat) at 32×32 resolution
- Trained for 12 hours on 20,000 images on Kaggle **P100 GPU (29GiB RAM, 16 GiB VRAM)**, strategically optimized the model architecture and training processes to overcome significant compute and memory constraints.

YOLO-based Smart Security Surveillance System Sep 2023 – Nov 2023
Freelance Project – Real-Time Threat Detection

- Engineered a real-time multi-camera surveillance system using **YOLOv11**, **InsightFace**, and **OpenCV** for object detection and facial recognition across **120,000+ images** (12+ GiB), enabling intelligent edge-based security monitoring.
- Trained a custom **YOLOv11n** model (**2.6M parameters**, **6.6 GFLOPs**) on **35,000+ labeled images** spanning **12 detection classes**, achieving **mAP50: 0.70** and **mAP50-95: 0.517** through iterative optimization and evaluation.
- Developed a **Flask-SocketIO** inference pipeline with real-time detection, unknown-face recognition, event logging, and Flutter-based alerting, achieving **10–15 FPS** deployment on **Jetson Nano** edge hardware for low-latency inference.

Academic Research Paper Assistant Nov 2024 - Nov 2024
Personal Project

- Developed an **Agentic GraphRAG system** leveraging 4 specialized tools and 4 intelligent agents to automate the search, retrieval, and intelligent querying of hundreds of academic papers, cutting manual research time by an estimated **40-50%**.
- Tools/Frameworks used: Crewai, Langchain, Ollama, llama3.1, arXiv API, Neo4j, Groq API

Achievements

1st Place	Innovate48 | Jaguar Land Rover ML/AI Hackathon, a college-wide 2-day Hackathon.	Sep 2023
4th Place	Kriti 2024 Inter Hostel AI/ML challenge, Research Paper Categorization	Feb 2024
1st Topper	Diploma in Electronics and Communication Engineering, Govt. Polytechic Barauni, Begusarai.	2020

Technical Skills

- **Programming Languages:** Python, C/C++, Java
- **Web Technologies:** FastAPI, Flask, Streamlit, Chainlit, SpringBoot
- **ML / AI Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Matplotlib
- **NLP / LLM Tools:** LangChain, HuggingFace, CrewAI, LiteLLM
- **Data / Databases:** MySQL, PostgreSQL, MongoDB, Milvus, Neo4j, Pandas, NumPy
- **DevOps/Infrastructure:** Docker, Git, AWS , Kubernetes, Kafka

Relevant Coursework

Computer Science: Data Structures and Algorithms, Database Management Systems (DBMS)
AI/ML: Machine Learning, Deep Learning, Recommendation System Design using Deep Learning, Applied Time Series Analysis